

Tanvir Ahmed Shanto

No Sponsorship Required | Open to Relocate

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SUMMARY

Mechanical Engineer and PhD researcher with 7+ years of experience in designing, prototyping, and validating hardware for advanced manufacturing and mechanical systems. Experienced in experimental design, failure analysis, finite element analysis (FEA) & multi-physics simulations.

TECHNICAL SKILLS

CAD:	AutoCAD, SolidWorks
CAE/Simulation:	Abaqus, ANSYS, HyperWorks, HyperStudy, Altair Inspire
Additive Manufacturing:	FFF, SLA, MJ, DLP, multi-axis extrusion, DFAM, LPBF
Data/Programming:	Python, MATLAB, Fortran, Polyworks, Statistical Analysis (DoE, Regression)
Material Characterization:	DSC, TGA, DMA, FTIR, OM, SEM, EDS, AFM, XRD, μ -CT
Standards/Methods:	ASME Y14.5, GD&T; ASTM, API, ISO test methods
Lean and Six Sigma:	Root Cause Analysis, DMAIC, SPC, Kaizen, 5-Whys, 5S, and 8D

WORK EXPERIENCE

Co-Founder & Technical Lead / Increscent, Inc. (Texas) *Aug'2022 – Present*

- Led commercialization planning for a novel 3D printing method to improve strength and dimensional accuracy, and repeatability in extrusion-based additive manufacturing.
- Led customer discovery and market validation activities; translated voice-of-customer (VOC) insights into product requirements, use cases, and qualification criteria.
- Produced technical documentation (pitch decks, test reports, and data packages) for fundraising events, business pitch competitions, and partner discussions.

Graduate Student Researcher / University of Texas Arlington *Aug'21 – Current*

- Designed and prototyped high-precision extruder hardware using GD&T and DFM/DFA principles; contributed to IP generation (one issued patent and one submitted patent application).
- Improved 3D-printed ABS mechanical performance by up to 68% through structured experimentation and process optimization; linked spatiotemporal thermal history to void formation to improve batch-print repeatability.
- Developed and executed test procedures for tensile, compression, 3-point bending, fracture, and fatigue testing.
- Built and validated Finite Element models & multi-physics simulations in Abaqus using experimental measurements and μ CT-informed geometry to explain fracture behavior and guide design/process changes.
- Performed materials and microstructure characterization and correlated results to mechanical behavior.
- Created supervised machine-learning models to predict mechanical performance from process parameters and thermal features, reducing trial-and-error in parameter selection.

Mechanical Engineer / Super Petrochemical Limited *Oct'18-Oct'19*

- Coordinated a 70-person mechanical workshop and maintenance crew by prioritizing work orders, allocating resources, and enforcing safe, quality repairs; applied Lean Six Sigma practices to support continuous operations.
- Reduced construction and maintenance costs of oil tankers by ~10% by identifying process waste and bottlenecks, improving planning and vendor coordination, and implementing continuous improvement.
- Designed process equipment and piping layouts using SolidWorks and AutoCAD Plant 3D and performed piping stress analysis (CAESAR II); partnered with fabrication/installation teams to improve manufacturability.
- Supported capital project development, including proposal review (component drawings, assembly MIs, bill of materials, analysis, etc.) for a \$10M mooring and pipeline program.

EDUCATION

PhD in Mechanical Engineering | The University of Texas at Arlington *Sep'21- Current*
BSC in Naval Architecture and Marine Engineering | BUET, Bangladesh *Feb'13-Sep'17*

AWARDS & PATENTS

- **NSF I-Corps Grant:** \$50,000 (customer discovery and commercialization training).
- **Patent:** Fused filament fabrication print-head system (US12496776B2).
- **Patent:** Fused filament fabrication adaptive annealing print-head system (Under filing)
- **Maverick Business Pitch Competition:** Secured \$35,000 by presenting an innovative business pitch, showcasing strategic planning and entrepreneurial skills.